



Need For Legal Compliance For Safe Solar Energy Waste Disposal

Strong legislative laws need to be framed by government authorities for mandatory recycling of solar energy system waste instead of landfilling, considering the massive 75,00,000 tons of waste that is likely to be generated until 2050. Solar module recycling is not yet commercially viable and is a net cost for the industry

Jaideep N. Malaviya

Solar energy is seen as a key element to address the issues of climate change. The compelling political commitments and declining costs are resulting in rapid market development from just 1GW of solar photovoltaics (PVs) in 2012 to 30GW in 2019 India. It has emerged as the third largest solar PV market in the world after China and the US with a cumulative installed capacity of over 28GW as of 2018. Fig. 1 shows the year-on-year market growth of solar PV installations and projections until the year 2030.

In solar thermal sector the cumulated solar water heater installed capacity is little over 14 million square metres but is now largely dominated by evacuated tube collectors.

The Indian government has set an ambitious target of achieving 100GW solar PV and 20 million square metres of solar thermal by March 2022. Further, the solar PV power installed is likely to touch 350GW by 2030 and ultimately 1,000GW by 2050.

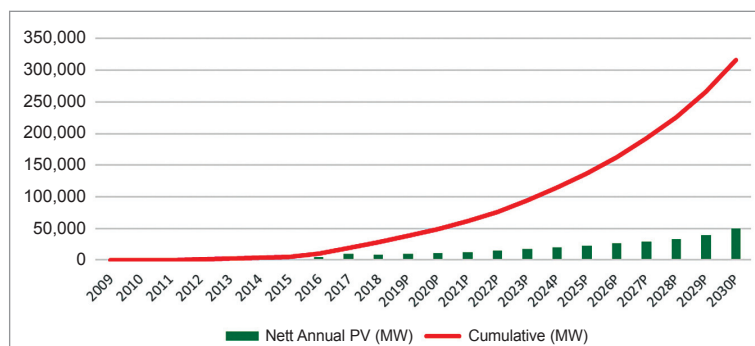


Fig. 1: Year-wise solar PV installed capacity and projected in MW in India

It is estimated that cumulative PV module volume will grow little over 8.5 million tons by 2025 and little under 20 million tonnes by 2030. With such robust growth to occur not much attention has so far been paid to module recycling in India.

The three principles of waste management are: reduce, reuse, and recycle. Owing

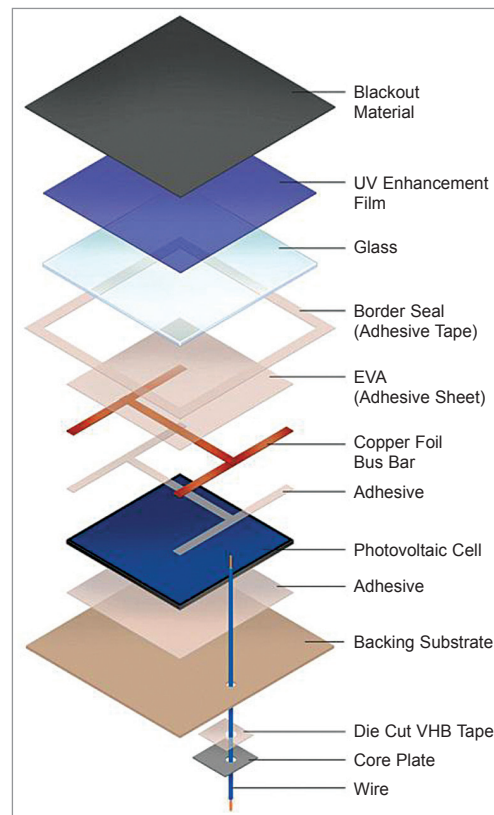


Fig. 2: Typical configuration of crystalline solar PV module